ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. IT ENGINEERING â€" SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) State Master's Theorem for divide-and-conquer recurrences. Solve T(n) = 4T(n/2) + n^2 and T(n) = 2T(n/2) + n log n. [7 Marks]
(b) Explain Strassen's Matrix Multiplication algorithm. Compare its time complexity with standard matrix multiplication. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) Explain amortized analysis techniques: Aggregate analysis, Accounting method, and Potential method with dynamic table expansion. [7 Marks]
(b) Describe Quick Sort using Randomized Pivot Selection. Prove expected running time is O(n log n). [7 Marks]

--- UNIT II ---

Q.2 (a) Explain Greedy choice property and optimal substructure. Solve Fractional Knapsack problem using Greedy strategy. [7 Marks]
(b) Describe Huffman Coding algorithm with prefix-free property. Analyze its complexity. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) Explain Job Sequencing with Deadlines algorithm using Disjoint Set Union (DSU). [7 Marks]
(b) Explain Minimum Spanning Tree algorithms (Kruskal and Prim) with cut property proofs. [7 Marks]

--- UNIT III ---

Q.3 (a) What is Dynamic Programming? Compare Dynamic Programming with Divide-and-Conquer and Greedy approaches. [7 Marks]
(b) Solve 0/1 Knapsack problem using dynamic programming with capacity W = 10 and given item weights and values. [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) Find Longest Common Subsequence (LCS) for sequences X = 'BACDB' and Y = 'BDCB' using DP table and trace back. [7 Marks]
(b) Explain Matrix Chain Multiplication algorithm. Find optimal parenthesization for chain <10, 30, 5, 60>. [7 Marks]

--- UNIT IV ---

Q.4 (a) Explain Backtracking technique. Solve 8-Queens problem using state-space tree. [7 Marks]
(b) Solve Subset Sum problem using backtracking. [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) Explain Branch and Bound technique. Solve 0/1 Knapsack problem using LCBB (Least Cost Branch and Bound). [7 Marks]
(b) Explain Traveling Salesperson Problem (TSP) using branch and bound with reduced cost matrix. [7 Marks]

--- UNIT V ---

Q.5 (a) Define Complexity Classes: P, NP, NP-Hard, and NP-Complete. What is Polynomial-time Reduction? [7 Marks]
(b) State and explain Cook's Theorem. Prove that Circuit-SAT / Boolean Satisfiability (SAT) is NP-Complete. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Prove that 3-SAT problem is NP-Complete by reduction from SAT. [7 Marks]
(b) Explain Vertex Cover and Hamiltonian Cycle problems and their NP-Completeness. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****